

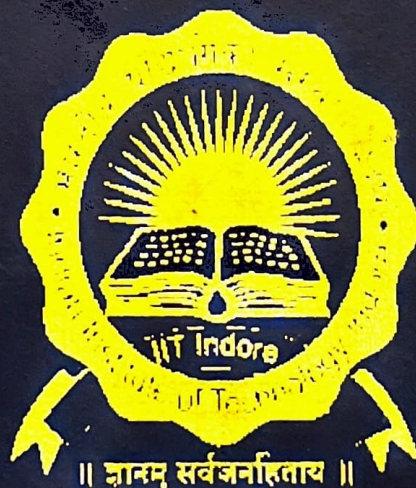
B.TECH. PROJECT REPORT
ON

FAKE NEWS DETECTION

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Chapter 1 - Introduction

People have a growing tendency to seek out and consume news from social media platforms rather than traditional news organisations. This is likely due to the fact that an increasing portion of our lives is spent communicating online through social media platforms. There are many reasons for this shift in consumption habits, including the nature of the social media platforms themselves. Traditional news media, such as newspapers, can be expensive and time-consuming to consume, while social media, such as Facebook and Twitter, can be more convenient and cost-effective for news consumption. Even though social media platforms come with a number of benefits, traditional news organisations continue to produce more accurate and reliable reporting than social media platforms. On the other hand, it is relatively inexpensive to offer news online, and it is much quicker and easier to transmit via social media.

In the past, the term "fake news" was used to refer to any type of unique material, including satires, hoaxes, news propaganda, and clickbait articles. In this project we went with the standard definition followed by many research paper i.e., Fake news is defined as **"The news articles that are intentionally and verifiably false, and could mislead the readers."**

Manual fact-checking is one approach for mitigating the major harmful impacts of false news. This incorporates fact-checking by experts as well as crowd-sourced manual fact-checking. Expert fact checkers can attain a high degree of accuracy, but this strategy is labour and time demanding, and scaling it to deal with the development of fake news is challenging. Although the crowd-sourced manual fact-checking technique is scalable, it will be assigned a label that is significantly less trustworthy since it does not match the criteria for reliable fake news identification. Automatic false news detection algorithms have been developed to overcome the limitations of manual fact-checking systems. The vast majority of previously conducted research has concentrated, for the most part, on identifying false information by analysing its linguistic characteristics. On the other hand, there is a phenomenon that can't be overlooked, and that is the fact that more and more news stories published on social media

platforms contain information presented in a variety of formats, such as text, photos, and videos.

The following is an explanation of why multimodal information should be used: (i) different modes help identifying different aspects of a story (ii) information gathered from multiple modalities complements each other in judging the veracity of a news article.

In this project, we mainly focus on the multimodal dimensionality of fake news and propose a novel classification model to detect fake news

Chapter 2 – Related work

2.1. Fake News Definition

False news, rumour, and disinformation are all phrases that are used interchangeably with fake news[1]. According to recent studies definition of fake news is “The news articles that are intentionally and verifiably false, and could mislead the readers.” [2].

2.2 Fake news detection

The objective of the task designed to detect false news is to determine whether or not the information being provided is authentic. The majority of currently available solutions frame the challenge of detecting false news as one of binary classification (fake or real). In certain circumstances, it may also be referred to as a multi-classification problem, a regression problem, or a clustering challenge. The literature on classification of fake news is quite extensive. In this section, we will present a quick overview of the work that has been done in the following categories: i) content-based ii) social context-based iii) propagation-based, and iv) multimodal fake news detection.

2.2.1. Content-based fake news detection

In the earlier investigations, text-based characteristics were obtained via linguistic features [5], [6]. Hand-crafted linguistic features for detecting fake news across multiple topics and contexts are challenging to generalise[7]. It's been demonstrated that an RNN-based model can learn the hidden representation of a temporal textual feature automatically, outperforming techniques that rely on hand-crafted features [8]. To capture the long-range interdependence among sequential information, soft-attention and RNN-based models are utilised to train selected temporal feature representations of articles[9].

The representation of textual characteristics can be enhanced by using adversarial learning between a text generator and a fake news classifier. Additionally, researchers investigated content-based fake news detection utilising a variety of methods such as user

response generation [11], text generation [10], reinforcement learning [12] and attention-residual network [13].

2.2.2. Social context based and Propagation based models

Textual characteristics aren't the only ones that can help differentiate fake news from real news; user demographics and propagation-based characteristics can also be used as auxiliary information. The relationship between user demographics and news credibility can be provided through systematic research [14]. Through the attention mechanism, Guo et al. combined the propagation based and user-based features with textual features [15]. Fake news, according to Vosoughi et al., spreads faster, and more widely on social networks than real news [16].

2.2.3. Multimodal fake news detection

In contrast to all of the preceding studies, visual information has also been used as supplemental information in order to determine the authenticity of news items [17], [18], [19]. Jin et al., who presented a methodical study comparing image characteristics of real and fake news [21]. Regardless, these features are still hand-crafted and do not collect information about more complicated visual content. Jin et al. suggested an RNN-based multimodal fusion false news detection system. They were motivated by DNN, which achieved outstanding results for the challenge of encoding text and image information [20]. The attention mechanism brings together the visual, as well as textual and social features. By eliminating text and visual event-specific information, Wang et al. proposed a multi-task learning model to learn textual and image feature representations across all posts [22]. Similarly, Memory networks were utilised by Zhang et al. in order to learn event-invariant features for the purpose of classifying fake news at event level. This resulted in increased generalizability for emerging new occurrences [23]. With the help of VAEs, a framework for learning shared latent representations across multiple media has been proposed by [24]. A study by Singh et al. finds that combining text and image features improves performance on the binary classification of fake news [25].

Chapter 3 - Dataset

3.1. Twitter Dataset

Twitter MediaEval dataset a multimodal dataset widely used in most of the previous works. Dataset derives from Verifying Multimedia Use task whose aim is to differentiate fake news and authentic news using an automatic method.

The dataset contains 17806 tweets accompanied with their associated images related to various events. A total 15629 samples were used for training purpose out of which 9404 samples were annotated as fake and 6225 samples as real. For testing purpose, we have 2177 samples in which 1186 samples are labelled as fake and 991 as real. The dataset is quite balanced.

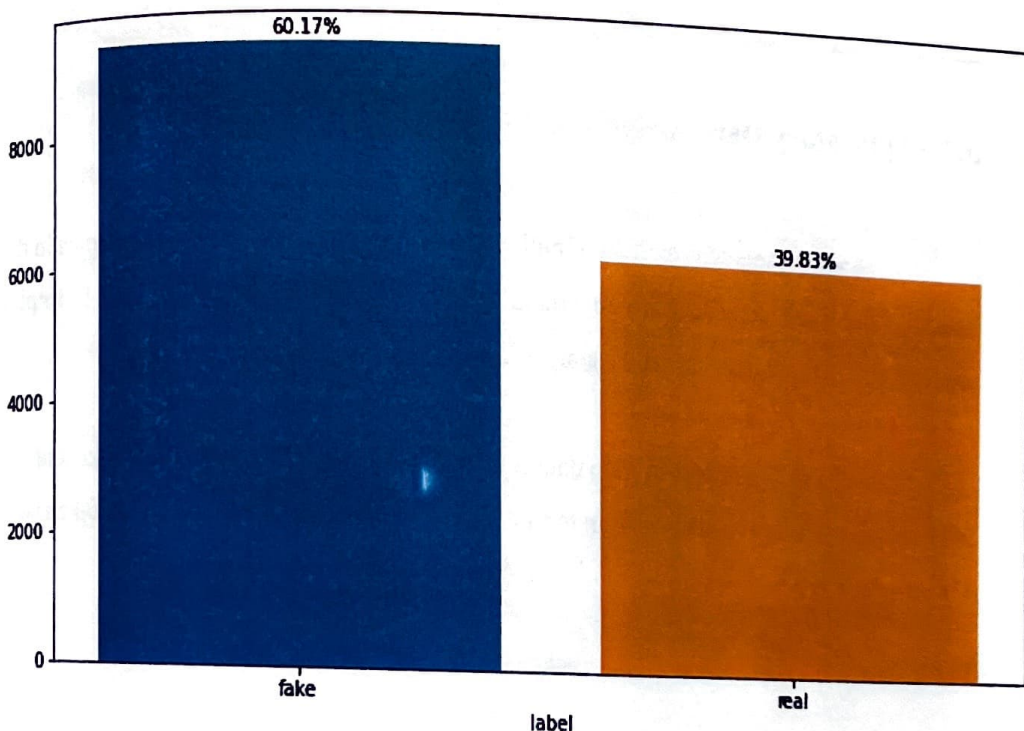


Figure 1: Percentage of #fake and #real instances in train data frame

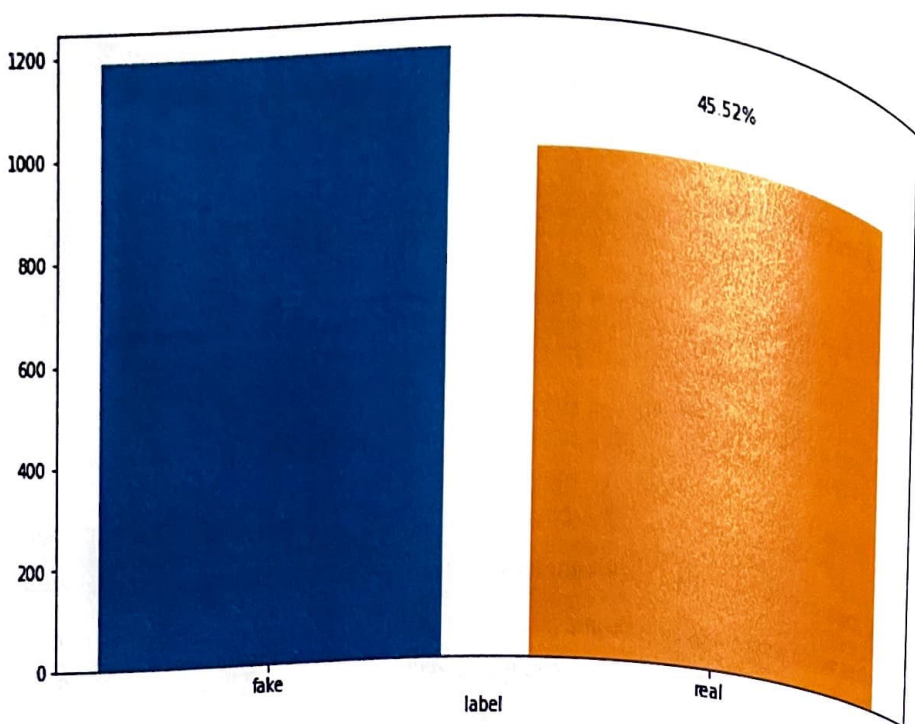


Figure 2: Percentage of #fake and #real instances in test data frame

3.2. Exploratory Data analysis (EDA)

EDA steps included removal of null values, removal of miscellaneous samples in the dataset. After these steps dataset size reduced to 15629 samples out of which 13241 samples are for training the model and 1098 samples for testing.

Out of 15629 samples in train data frame 7324 samples belongs to class fake and 5917 samples belongs to class real. In test data frame 630 samples are labelled as fake and 438 samples as real.

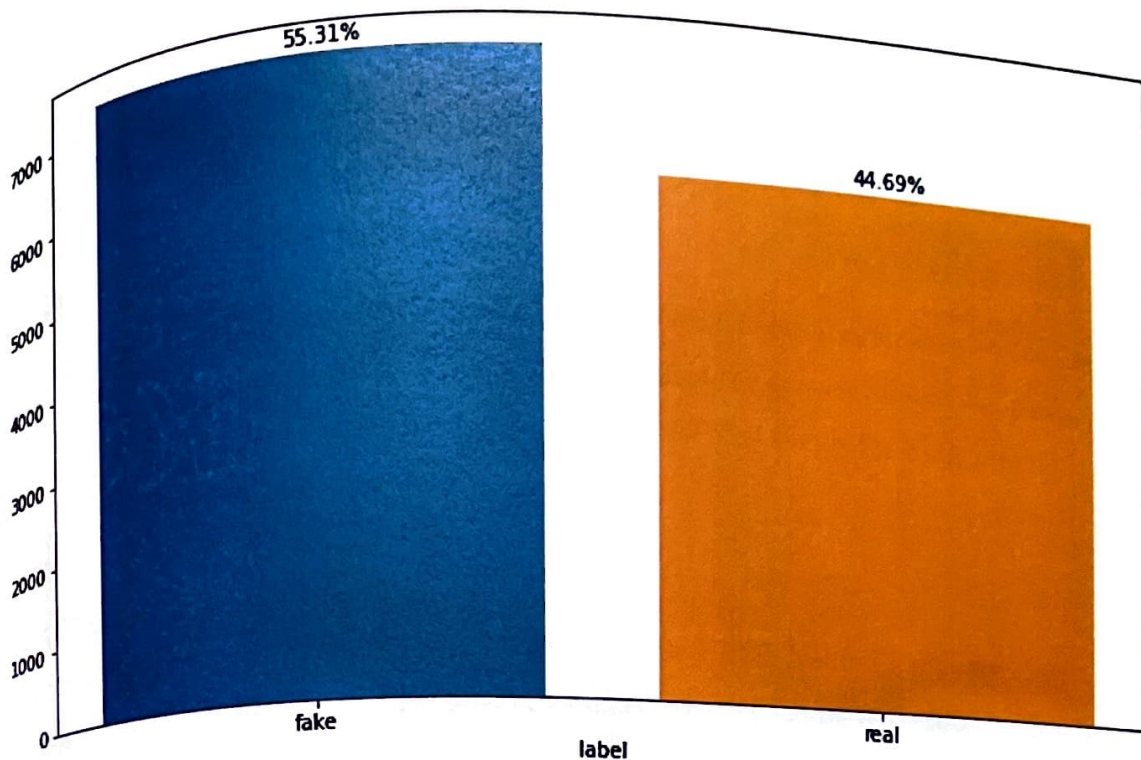


Figure 3: Percentage of #fake and #real instances after EDA, in train data frame

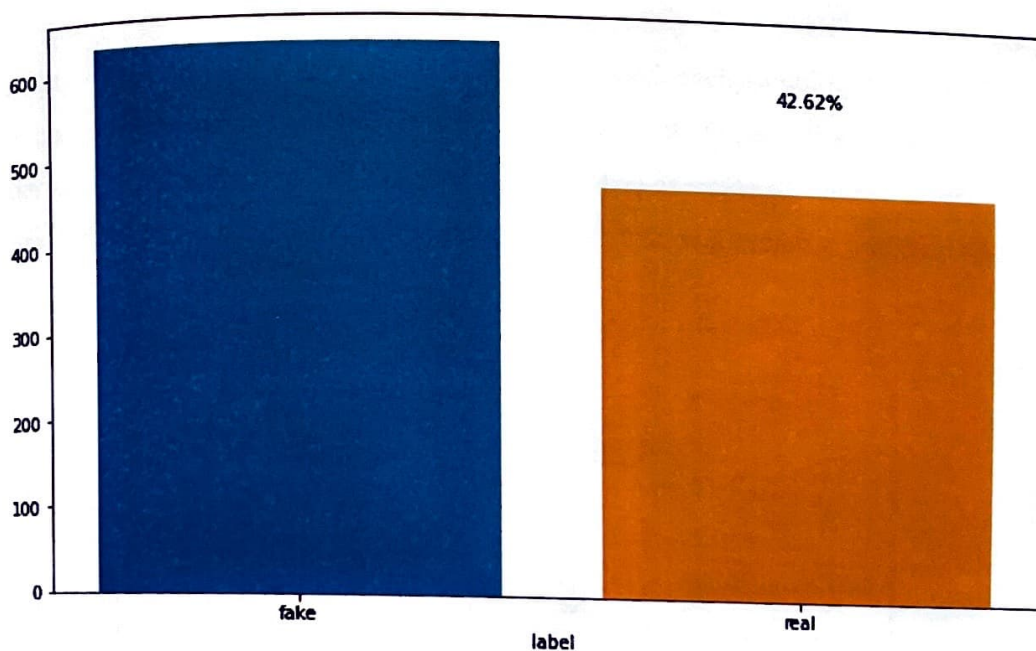


Figure 4: Percentage of #fake and #real instances after EDA, in test data

Chapter 4 - Methodology

4.1 Textual Feature Extraction

Consider this as a sub-model of the multimodal framework that we employ for the task. This is responsible of extracting textual features from the dataset content viz. posts. Here we make use of state-of-the-art model namely BERT that stands for Bidirectional Representations from Transformers.

4.1.1 BERT

In 2018, researched at Google Research proposed the BERT Natural Language Processing Model. It was proposed to attain cutting edge-precision on numerous language modelling tasks, such as

- Language Comprehension Test
- Situation with Adversarial Generations

After that, the BERT base and BERT large were released, each including a different number of encoders (12 or 24). Simply said, BERT is an encoding stack that is used by transformer architecture. An encoder-decoder network is an example of a transformer architecture. Self-attention was used on the encoder side, whereas attention was used on decoder side.

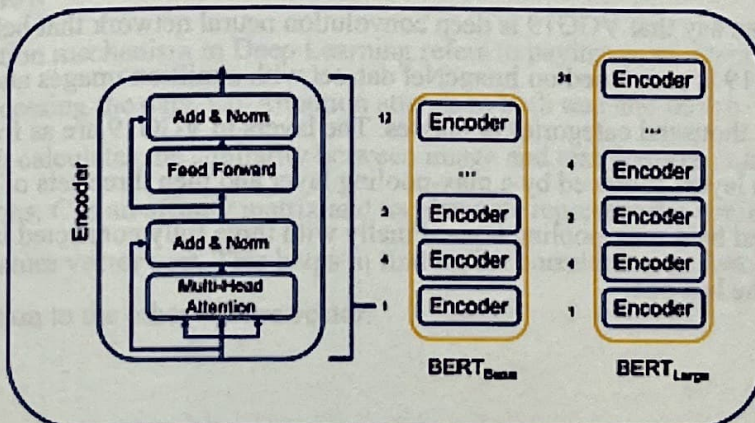


Figure 5: BERT general structure¹

¹ Source: <https://www.analyticsvidhya.com/blog/2021/12/fine-tune-bert-model-for-sentiment-analysis-in-google-colab/>

Attention is paying greater attention to certain factors while processing the data. BERT takes three inputs viz. token embeddings, segment embeddings and positional embeddings. Further topology includes larger feedforward-networks (with 768 for BERT base).

Now BERT is pre-trained with two unsupervised tasks which are

- Masked Language Modelling (MLM)
- Next Sentence Prediction (NSP)

This makes BERT understand the language better and hence give better contextual word embeddings requisite for better performance of model. We use this model to extract textual features from the samples by passing the contextual word embedding which is the output of BERT through a dense layer to reduce it to dimension of length 32.

4.2 Image Feature Extraction

AlexNet was released in 2012, and it improved on traditional Convolutional neural networks. We can think of VGG as a successor to AlexNet and it was developed by a group at Oxford called the Visual Geometry Group (VGG). It carries and improves on ideas from its predecessors, and it uses deep Convolutional neural layers to improve accuracy.

4.2.1 VGG19

We can say that VGG19 is deep convolution neural network that helps in classifying images. VGG19 is pretrained on ImageNet dataset with a million images and classifying them across a thousand categories or classes. The layers in VGG19 are as follows: two sets of 3 convolution layers followed by a max-pooling layer and then three sets of 4 convolution layers followed by a max-pooling layer. Finally with three fully connected layers and a SoftMax on the last one.

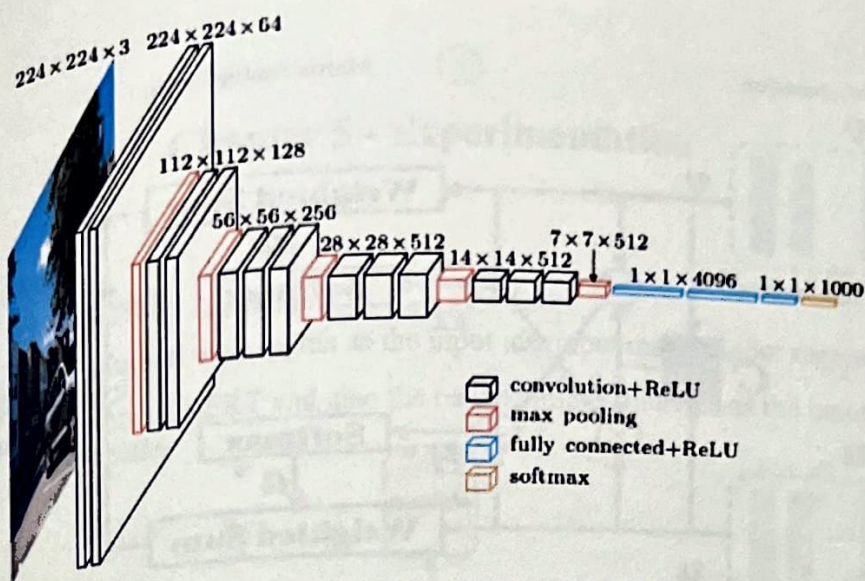


Figure 6: Architecture of VGG19²

VGG19 takes an RGB image size of (224×224) , that implies a matrix of $(224 \times 224 \times 3)$. The kernels used are of (3×3) size and stride of 1 pixel is employed to cover the whole notion of image. Max pooling is done over (2×2) pixel window size with stride 2. We parse the images through VGG19 and extract the output from before last layer as the last one is for classification task. Finally, we take the output from VGG19 and pass it through a dense layer to reduce it to dimension of length 32.

4.3 Co-Attention

The attention mechanism in Deep Learning refers to paying great attention to certain factors while processing the data. Co-Attention attends to both text and image simultaneously. It calculates the similarity between image and text features at all pairs of text and image locations. C is an affinity matrix and its elements represent the similarity between corresponding feature vector pair. This helps in finding the correlations or features that are important in relation to the other feature vector.

² Source: <https://arxiv.org/pdf/1409.1556v6.pdf>

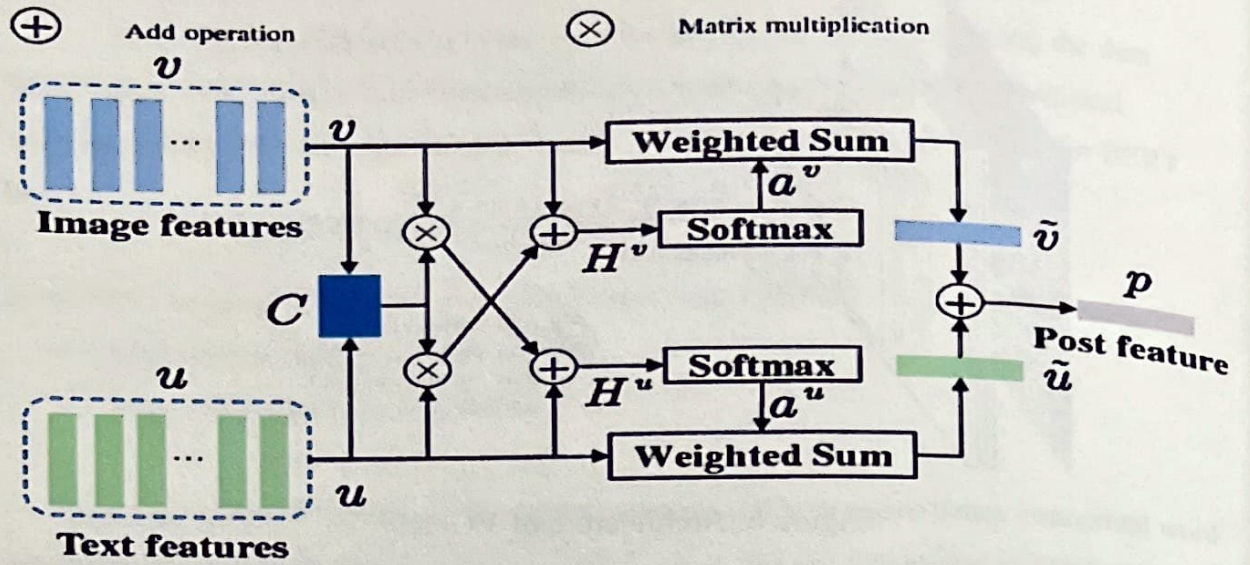


Figure 7: Co-Attention Architecture³

Based on the affinity matrix, we next transfer the image and text features for each other to further exploit the correlations between image and text features. This is done through multiplying both u and v with C . Then we add it to the weighted matrix of opposite feature represented by the criss-cross in the Fig 7. After getting the attention-weights, obtained by applying SoftMax function to the computed vectors, we do the weighted sum. Finally, we output the post feature which is a concatenation of the obtained attention-weight vectors.

4.4 User Feature Extraction

Detecting fake news based on news content is difficult and non-trivial to detect. As a result, we'll need some additional information. Because different sources alter different modalities and different modalities complement each other in identifying authenticity, it would be beneficial to combine multimodal information with user attributes. After performing EDA, as explained in [section 3.2](#) like removing NULL values and applying one-hot encoding, we pass it through a fully connected layer to get a feature vector of dimension size 32.

³ Source: <https://ojs.aaai.org/index.php/AAAI/article/download/4528/4406>

Chapter 5 - Experimentation

5.1 Base Model

We prepare a model with inputs as the input ids, input masks, input segments that correspond to the inputs of BERT and also the resized image matrices as the input to VGG19.

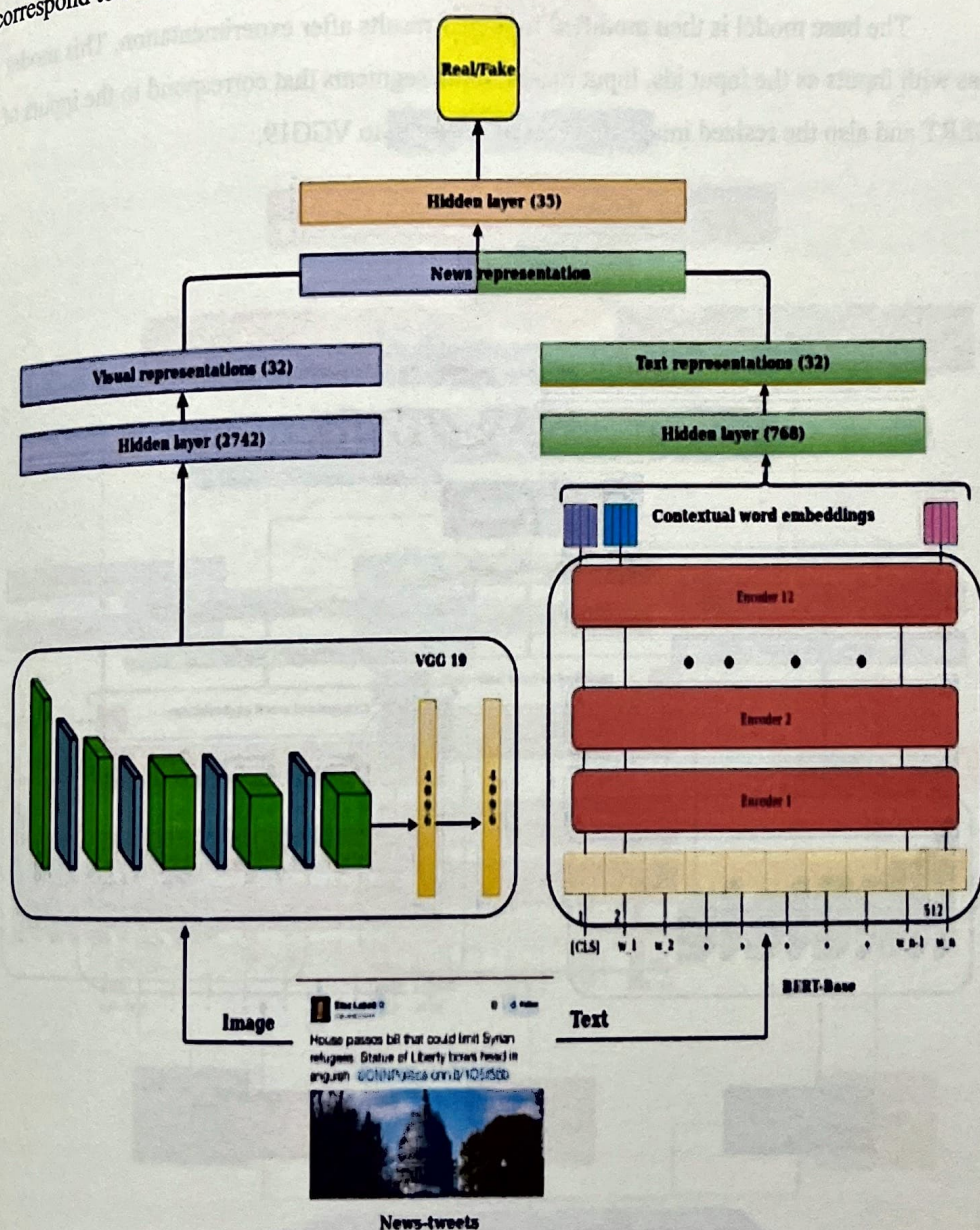


Figure 8: A schematic diagram of the Base Model. Number of neurons in each layer is mentioned in ()

The textual and visual representations that are obtained from BERT and VGG19 respectively are then simply concatenated to obtain a news representation vector of dimension size 64. This is then passed through a hidden dense layer to finally classify into fake or real by the model. This we term as the Base Model.

5.2 Base Model with Co-Attention

The base model is then modified for better results after experimentation. This model has with inputs as the input ids, input masks, input segments that correspond to the inputs of BERT and also the resized image matrices as the input to VGG19.

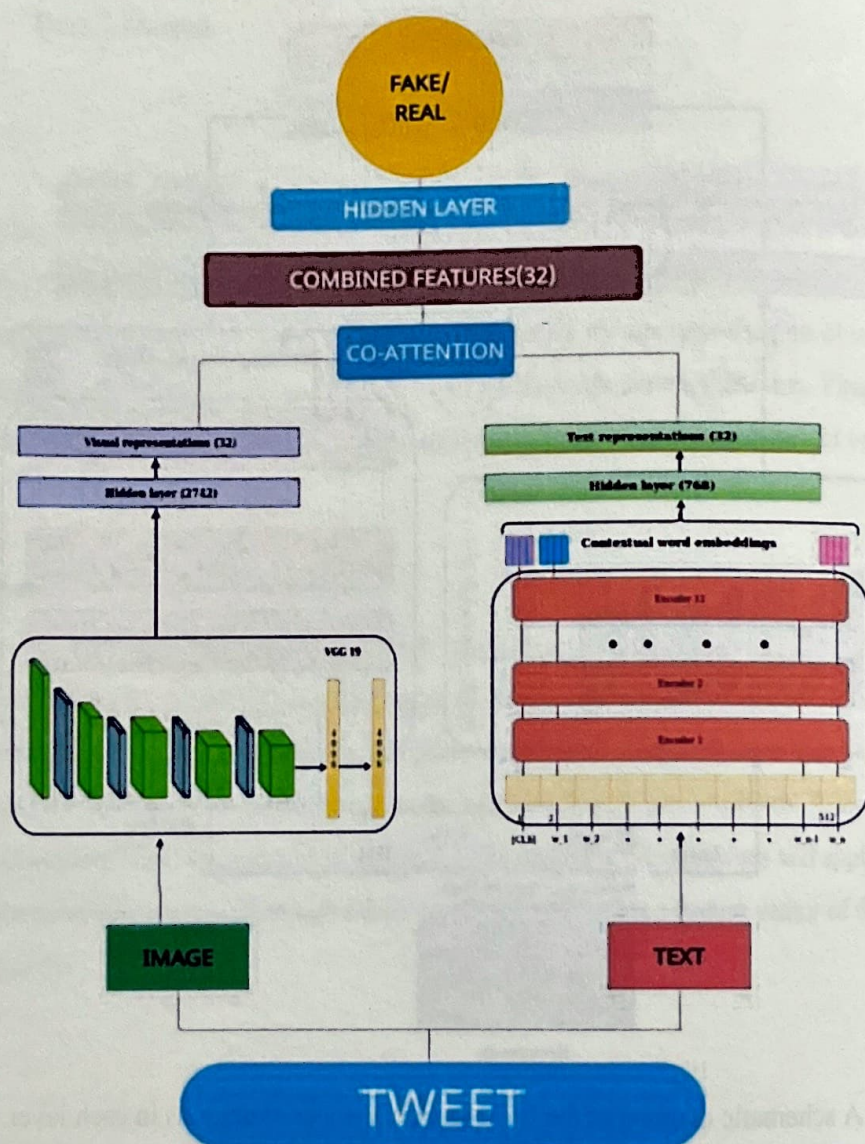


Figure 9: A schematic diagram of the Base Model with Co-Attention

We now apply co-attention on the obtained textual and visual representations. This will result in combined feature news representation, a vector of dimension size 32. This is then passed through a fully connected hidden layer to finally classify fake or real.

5.3 Proposed Model

The model takes inputs as the input ids, input masks, input segments that correspond to the inputs of BERT and also the resized image matrices as the input to VGG19.

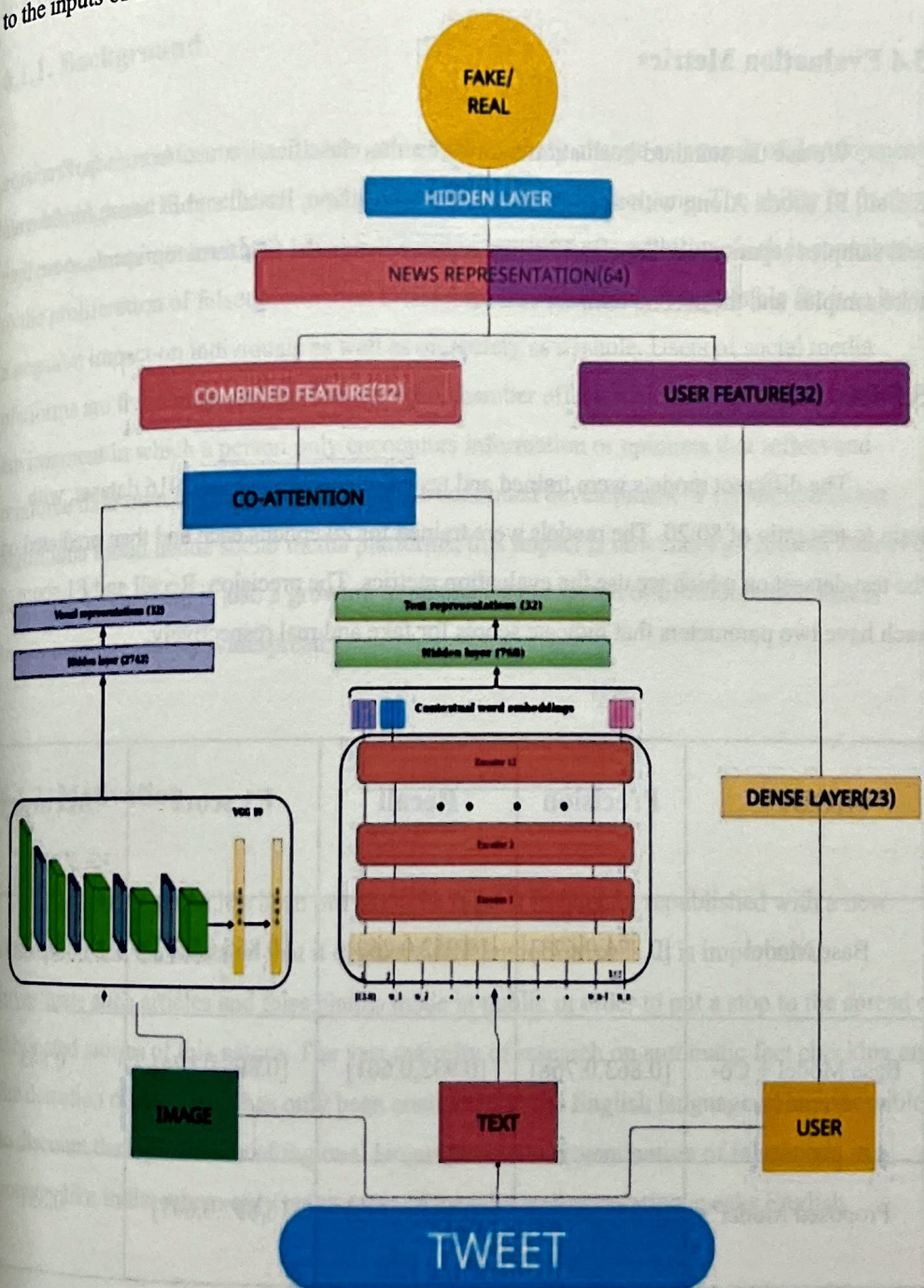


Figure 10: A schematic diagram of the Proposed Model

After we get the textual and visual representations, we apply co-attention on these to get a combined feature vector of dimension size 32. At the same time, we also get user features of dimension size 32 by passing the user demographics through a dense layer. Now, we concatenate combined features with user features to get the news representation vector of dimension size 64. This will be finally passed through a hidden fully connected layer to classify fake or real.

5.4 Evaluation Metrics

We use the standard evaluation metrics for this classification viz. Accuracy, Precision, Recall, F1 score. Along with this, we calculate the Precision, Recall and F1 score for fake and real samples separately using classification report wherein the first term represents score for fake samples and the second term for real samples.

5.5 Results

The different models were trained and tested on the MediaEval 2016 dataset, with train to test ratio of 80:20. The models were trained for 20 epochs each and then predicted on the test dataset on which we use the evaluation metrics. The precision, Recall and F1 score each have two parameters that indicate scores for fake and real respectively.

MODEL	Precision	Recall	F1 score	Accuracy
Base Model	[0.776,0.673]	[0.952,0.262]	[0.855,0.377]	0.7659
Base Model + Co-Attention	[0.863,0.768]	[0.932,0.601]	[0.896,0.674]	0.843
Proposed Model	[0.857,0.733]	[0.931,0.573]	[0.892,0.647]	0.851

Table 1: Performance comparison of Base Model with Proposed Model

Chapter 6 – Work in Indic languages

6.1 Introduction

6.1.1. Background

Real-time information is undergoing revolutionary change as a result of developments in technology and the increased availability of the internet to everyone. The ability to freely express one's opinions without being subjected to any kind of credibility check is contributing to the proliferation of false information throughout the ecosystem. It is possible for it to have a negative impact on individuals as well as on society as a whole. Users of social media platforms are frequently subjected to the echo chamber effect, which can be defined as an environment in which a person only encounters information or opinions that reflect and reinforce their own beliefs. As a result of the continued development of recommendation algorithms found inside social media platforms, this impact is now more prominent than ever before. In India, there is also a growing trend toward the spread of disinformation that is becoming increasingly widespread.

6.1.2. Motivation

Information that has been proven to be false is frequently republished with a new description and the assertion that it depicts a different occurrence. It is important to investigate such articles and false claims made in public in order to put a stop to the spread of fabricated stories of this nature. The vast majority of research on automatic fact checking and the detection of fake news has only been conducted in the English language. It is impossible to discount the significance of regional languages in the dissemination of falsehoods in a country like India, where only ten percent of the educated population speaks English.

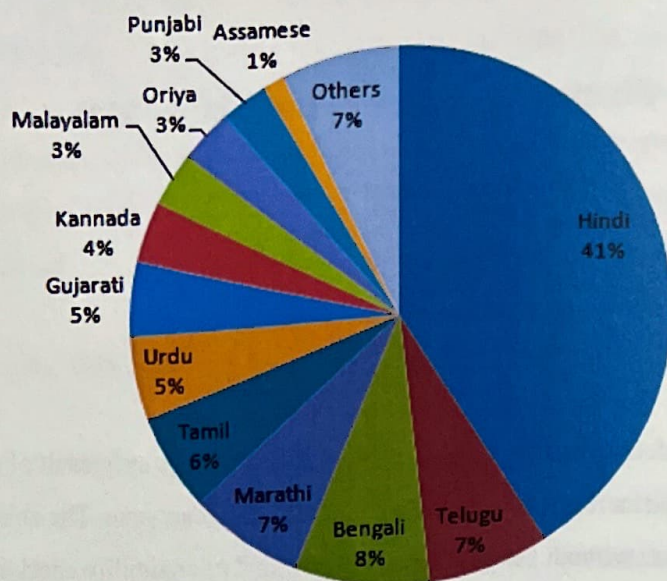


Figure 11: Percentage distribution of languages spoken in India

The challenges in finding a balanced multilingual multimodal dataset in low-resource Indic languages is un-paralleled to any, reasons being: i) lack of multimodal information, ii) a smaller number of samples in dataset, and iii) a smaller number of languages covered.

In this project we propose a first large-scale multilingual multimodal dataset on low-resource Indic languages to the best of our knowledge and build a robust model for fake news detection.

6.2. Related work

Less research on fake news detection has been on low-resource Indic languages. Due to lack of a publicly available benchmark dataset, automatic fact-checking will be quite difficult. Sharma et al. recently proposed the IFND, which stands for the Indian Fake News Dataset. The dataset contains 37,809 real news samples and 7,271 fake news samples, respectively [32]. Dhawan et al. put forward the idea of conducting research under the name FakeNewsIndia to investigate instances of fake news in India. The group selected 4,803 stories to use in their compilation [33].

Despite the fact that the datasets have made an effort to develop resources that are tailored to the Indian region, it is still subject to a number of restrictions. I The IFND dataset has a significant degree of imbalance. The veracity of the sources is not guaranteed in any way, (ii) the number of samples included in the FakeNewsIndia dataset is relatively small. (iii) Both of the curated datasets only contain samples written in English, and they are lacking data written in regional languages. (iv) A website can have a wide variety of characteristics, but the focus of these papers will be narrowed down to a select few aspects. It's possible that this will result in a loss of information.

To overcome this limitations Singhal et al. put forth a dataset called FactDril a multilingual multimodal dataset [28].

2.2 Data Extraction

Chapter 7: Dataset Curation

7.1 FactDril

FactDril stands for **Fact-Checking Dataset for Regional Indian Languages** is a multilingual multimodal dataset on low resource Indic languages. It contains 22435 samples of fake news extracted from eleven fact-checking websites. Dataset covers 13 languages as a whole out of which 11 are low-resource Indic languages and the rest two being English and Hindi.

Dataset contains 9058 samples of English language, 5155 samples of Hindi and the rest 8222 samples are distributed over various other languages (i.e Telugu, Tamil, Malayalam, Marathi, Urdu, Odia, Punjabi, Bengali, Gujrati, Burmese and lastly Sinhala). The dataset follows detailed characterization of three M's: i) Multilingual, ii) Multimodal and iii) Multidomain.

The features that were curated from the dataset can be further divided into the categories of meta, textual, media, social, event, and author features.

The main motive behind FactDril is that to understand the dynamics of fake news. Although the dataset is unique on its own it comes with some limitations which are i) all the samples in the dataset are labelled as fake implies the dataset is biased on one side ii) so a classification task is not possible to implement on FactDril.

7.2 Data Extraction

To overcome the limitation mentioned above, we started extracting real news samples from twitter accounts across various trusted news channels. For extracting tweets, we used a

social networking scraper called **Snsrape** over the time period 2013-2022 and collected 32,000 real news samples across 13 languages mentioned in FactDril dataset.

Language	Username	#samples collected
English	the_hindu	10,000
Hindi	aajtak	5000
Telugu	TV9Telugu	1500
Tamil	polimernews	1800
Malayalam	24onlive	1700
Marathi	TV9Marathi	1700
Bengali	News18Bengali	1500
Gujarati	abpasmitatv	1500
Urdu	News18Urdu	1100
Punjabi	abpsanjha	900
Odia	OdishaReporter	800
Burmese	Voaburmese	700
Sinhala	SLNewsSinhala	800

Table 2: Shows #real news samples collected in each language and their respective usernames of twitter accounts

7.3 Data Pre-Processing

Pre-Processing steps in the dataset include:

- Removal of tweets that doesn't have any article link
- Removal of NULL values
- Removal of expired image links in the dataset.

Before		After	
54435		32666	
Fake	Real	Fake	Real
22435	32000	16885	15781

Table 3: Shows total #each class samples in the dataset before & after pre-processing

Handling such huge number of samples is an arduous task so, we can make it easier by storing each language content with image downloaded in a separate folder in order to access faster and easier.

Language	Fake	Real
Telugu	799	1500
Marathi	1108	1700
Malayalam	1251	1700
Tamil	1314	1800
Bengali	956	1500
Gujarati	1003	1500
Punjabi	428	900
Odia	287	800
Urdu	596	1100
Burmese	128	700
Sinhala	256	800

Table 4: Shows #fake and #real samples in each language before pre-processing

Language	Fake	Real
Telugu	799	762
Marathi	658	882
Malayalam	1096	1408
Tamil	769	1081
Bengali	660	1387
Gujarati	974	1254
Punjabi	425	808
Odia	287	459
Urdu	529	739
Burmese	128	336
Sinhala	221	252

Table 5: Shows #fake and #real samples in each language after pre-processing

7.4 Dataset Statistics

Total number of samples present in dataset after pre-processing is 32666 out of which 16885 (51.69%) belongs to class fake and 15781 (48.31%) belongs to class real.

Dataset contains attributes such as content, image_url, image_id, label.

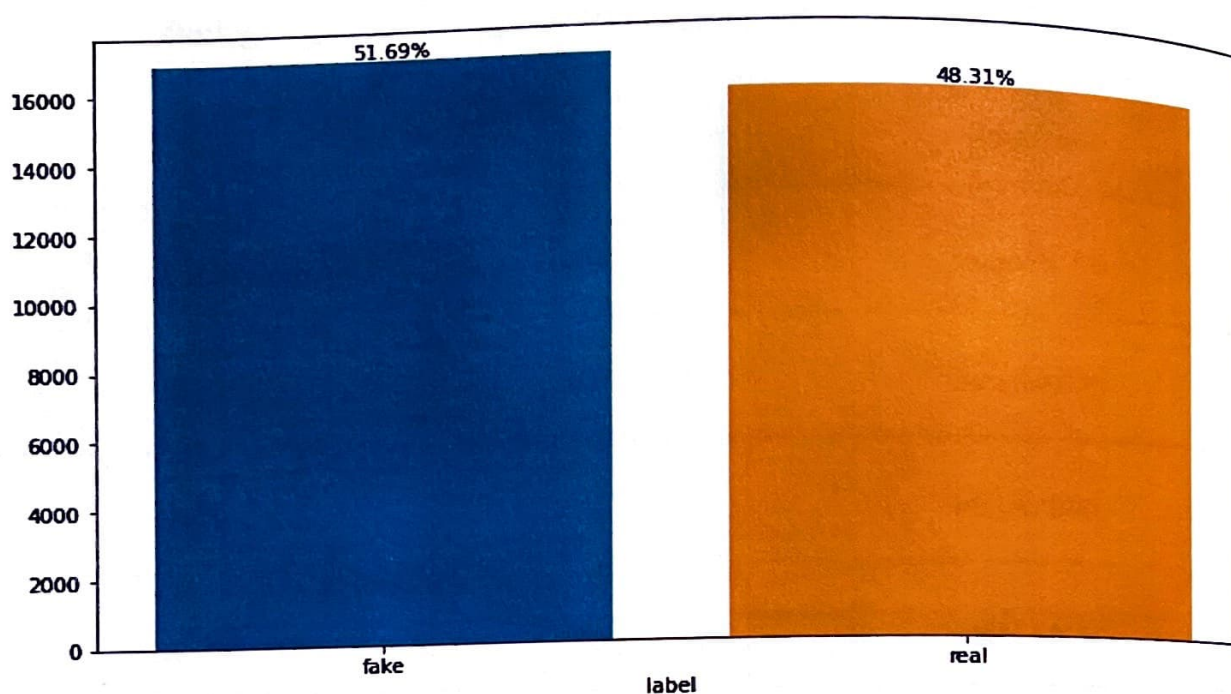
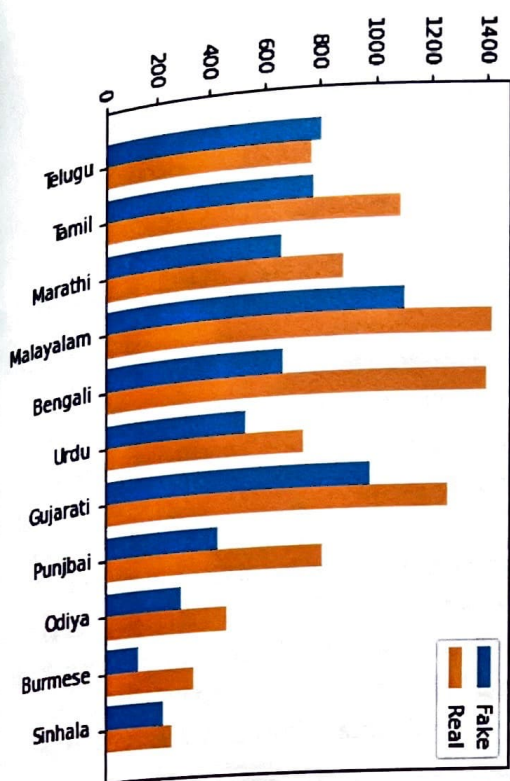
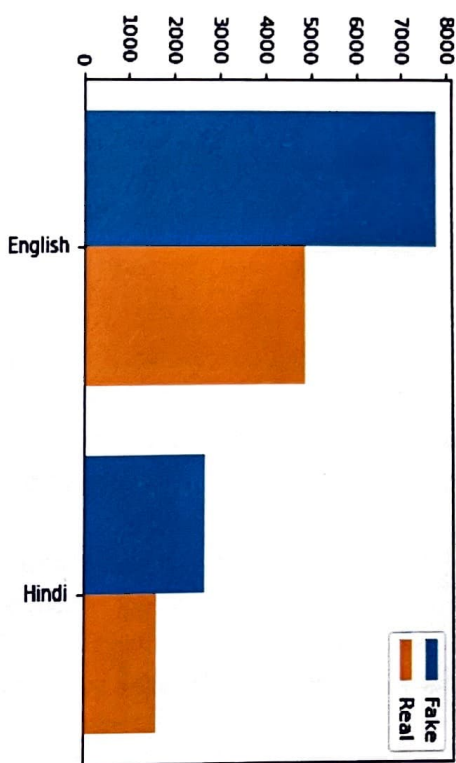


Figure 12: Percentage of #fake and #real class in the dataset

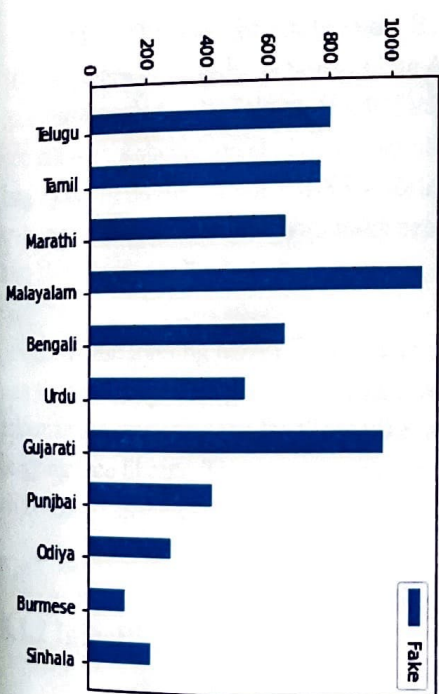


(a)

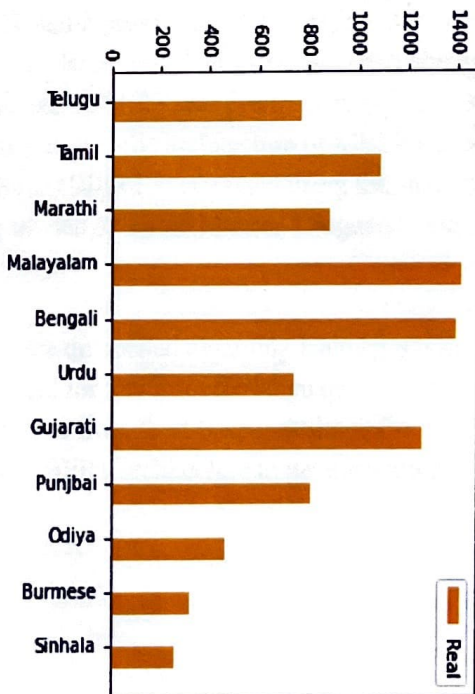


(b)

Figure 13: (a) Shows distribution of #real and #fake in each language whereas, (b) in English and Hindi



(a)



(b)

Figure 14: (a) Shows distribution of #fake whereas, (b) shows distribution of # real in each language

Chapter 8: Methodology

8.1 Textual Feature Extraction

Firstly, we need to extract text content from the tweets that were scraped. We use URL retrieve function in order to get the URL of the embedded article. Then we use the Triflatura package to get the content. This package stands out from others as it is compatible even when we have low resource Indic Languages in the data.

Then we truncate the obtained text content to a maximum size of 400 characters as some of the articles are way too long. Now, for extracting the textual features from the content, we employ state-of-the-art models such as MBERT, IndicBERT and MURIL. The output is then passed through a layer of LSTM units and then a dense layer to get the textual features and reduce the dimension size 32.

8.1.1 MBERT

MBERT stands for Multilingual BERT and it provides sentence representations for 104 languages. It includes languages like Arabic, Bosnian, Chinese and also Indic languages like Burmese, Bengali, Telugu, Hindi, Sinhala etc. MBERT was pre-trained on Wikipedia content with a shared vocabulary across all languages. The architecture of MBERT model is largely based on the original BERT architecture. MBERT was trained using the same pre-training technique as BERT with tasks being trained on as the Masked Language Modelling and Next Sentence Prediction.

While training MBERT, the samples were unlabelled regarding languages ensuring one shot learning takes place that would be useful for low resource languages. Firstly, we tokenize the text using the BertTokenizer available from 'bert-base-multilingual-cased' hugging face library. Then give it input to the MBERT architecture to get contextual embeddings as the output.

8.1.2 IndicBERT

Indic BERT is trained similar to MBERT and it is trained mainly on 11 Indian languages other than English. The languages are Punjabi, Hindi, Bengali, Odia, Assamese, Gujarati, Marathi, Kannada, Telugu, Malayalam and Tamil.

IndicBERT is almost same as MBERT but we can say that it is unique and exclusively for Indic languages. Tokenizer is imported with the help of AutoTokenizer module from 'indic-bert'. This is used to tokenize the input which is taken as input.

8.1.3 MURIL

MURIL is short for Multilingual Representations for Indian Languages. It supports 17 languages including English. The languages are Assamese, Bengali, Gujarati, Hindi, Kannada, Kashmiri, Malayalam, Marathi, Nepali, Oriya, Punjabi, Sanskrit, Sindhi, Tamil, Telugu and Urdu.

MURIL is trained on significantly large amounts of Indian text corpora only. It is pre-trained on mainly two tasks namely Masked Language Modelling, Translation Language Modelling. The Translation Language Modelling was basically done on two corpus that is on the Dakshina dataset and on the Indic-trans library.

MURIL crucially aims at handling challenges related to Indian Languages that include spelling variations, Transliteration and so on. The training of MURIL is similar to that of BERT with the addition of two tasks viz. Translation and Transliteration. Tokenizer is imported from 'google/muril-base-cased' with the help of AutoTokenizer. This tokenizer is used to tokenize the content and the parsed into MURIL to get the output.

8.2 Image Feature Extraction

The images are an important mode when we consider a multimodal approach. As such, extracting the main image from the article that has multiple images in it. So, we used an open-source software called Article Parser⁴ to get the main image link from the article.

After getting the image, we resize it and pass it through VGG19 and extract the output from the before last layer as the image features. The output is then passed through a dense layer to get feature vector of dimension 32.

8.3 Multimodal Fusion

We combine the textual and visual features after we have obtained them. Multimodal fusion aids in the classification of data greatly. We kept the train to test ratio as 80:20 of 32666 total number of sample.

⁴ <https://demos.pwshub.com/article-parser>

Chapter 9: Experimentation

We tested the dataset curated on different state-of-the-art models. Having a multimodal approach with each of them made us better understand as to which performs better for our case.

9.1 With MBERT

For this model, we employ MBERT architecture and after getting the textual feature from this, we concatenate it with the image features to get a combined news representation vector of dimension size 32.

9.2 With IndicBERT

For this model, we employ MBERT architecture and after getting the textual feature from this, we concatenate it with the image features to get a combined news representation vector of dimension size 32.

9.3 With MURIL

For this model, we employ MBERT architecture and after getting the textual feature from this, we concatenate it with the image features to get a combined news representation vector of dimension size 32.

9.4 Evaluation Metrics

We employ the standard evaluation metrics as mentioned earlier. In addition to it, we also calculated the overall Precision, Recall and F1 scores for fake and real samples separately too. The results of different models for five epochs are mentioned below.

9.5 Results

MODEL	Precision	Recall	F1 score	Accuracy
MBERT+ VGG19	[0.89,0.91], 0.90	[0.94,0.77], 0.88	[0.88,0.84], 0.83	0.80
IndicBERT + VGG19	[0.78,0.99], 0.98	[0.99,0.74], 0.74	[0.87, 0.85], 0.84	0.86
MURIL + VGG19	[0.83,0.95], 0.94	[0.95,0.82], 0.81	[0.89,0.88], 0.87	0.883

Table 6: Classification results on the dataset using MBERT, IndicBERT and MURIL models.

Chapter 10: Conclusion & Future Work

In the past, approaches to detecting fake news included natural language processing, knowledge graphs, computer vision, and user profiling. A multimodal technique has been shown to be necessary for trustworthy results. So, we propose a novel classification model on the benchmark dataset (MediaEval 2016), that outperforms the existing baseline model by a margin of 8% accuracy.

Also, we devised a Multilingual, Multimodal and Multidomain dataset that is based mainly on Indic languages. This is the first large scale balanced multilingual dataset on fake news detection over low resource Indic languages, to the best of our knowledge.

In the future, the architecture might be enhanced by implementing Optical Character Recognition (OCR) model. OCR helps recognize the text in the image. So, addition of auxiliary features might improve the performance of the model.

The dataset that we devised can be used for many use cases. Not only to classify, it can also be used to understand the dynamics of the content of fake or real news. As such, we hope this dataset would help accelerate NLP research for Indic Languages.

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B.TECH. PROJECT REPORT
ON

FAKE NEWS DETECTION

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DISCIPLINE OF COMPUTER SCIENCE AND ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY INDORE

May 2022

FAKE NEWS DETECTION

A PROJECT REPORT

*Submitted in partial fulfilment of the
requirements for the award of the degrees*

of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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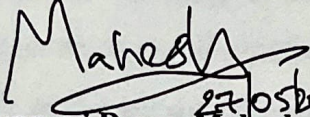
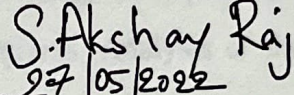
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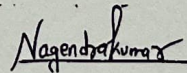
We hereby declare that the project entitled "FAKE NEWS DETECTION" submitted in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering completed under the supervision of **Dr. Nagendra Kumar, Associate Professor, IIT Indore** is an authentic work.

Further, we declare that we have not submitted this work for the award of any other degree elsewhere.

 
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CERTIFICATE by BTP Guide

It is certified that the above statement made by the students is correct to the best of my/our knowledge.



Dr. Nagendra Kumar

Assistant Professor

27-05-2022

Signature of BTP Guide

Preface

This report on “Fake News Detection” is prepared under the guidance of Dr. Nagendra Kumar, Assistant Professor, IIT Indore.

Through this report, we have tried to provide a detailed description of our approach, design and implementation of an innovative method to perform classification task on Fake News Detection. We have also proposed a novel model on Multilingual Multimodal dataset

We have tried to the best of our abilities and knowledge to explain the content in a simplified manner by using illustrative diagrams. Analysis of the results and architecture of the model are also discussed.

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Abstract

An alarming rise in the number of false news stories is a major problem in our culture. The wide dissemination of false information has the potential to have tremendously detrimental effects, both on individuals and on society as a whole. Therefore, the detection of fake news on social media has recently become an emergent field of research that is receiving a great deal of attention.

Image editing, text editing, audio editing, and video editing are the typical methods used to produce it. This suggests that there is a requirement for a multimodal system for the detection of fake news. In this project we propose a novel classification model which outperforms the existing baseline models.

Fake news is spreading like wildfire across the country, too. Most of the research on detecting fake news has been limited to English language only. Regionalism's significance in disseminating misinformation cannot be underestimated, especially in a country like India where less than 10% of the population is literate.

In this project we propose a first large-scale balanced multilingual dataset on low-resource Indic languages to the best of our knowledge.

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